

Sushant S. Mahajan

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SUMMARY

Research scientist and computational physicist building data-driven systems for solar physics, scientific inference, and time-series analysis. Experience spans observational data products, large-scale signal extraction, forecasting workflows, reproducible software, and inter-disciplinary research communication.

EXPERIENCE

Research Scientist, W. W. Hansen Experimental Physics Laboratory, Stanford University 2024–present

- Planned and lead the development of agentic AI-powered JSOC data-commons for data query and visualization.
- Lead research on solar magneto-hydrodynamics and large-scale flows from HMI/SDO observations.
- Produced new metadata and analysis workflows for solar magnetic-field and velocity-field data.

Postdoctoral Fellow, W. W. Hansen Experimental Physics Laboratory, Stanford University 2022–2024

- Published first-author work on differential rotation, torsional oscillation, and meridional flow.
- Helped deliver the Stanford Solar Observatories Group data-sharing effort recognized in 2026.
- Collaborated across research, software, and data management teams on mission-facing analysis.

Solar Physics Postdoctoral Fellow, Institute for Astronomy, University of Hawai'i 2019–2022

- Built analysis pipelines for active-region inflows, solar-cycle trends, and helioseismic flow inference.
- Developed a proposal for extracting new metadata from solar images for improving solar flare forecasting.

SELECTED IMPACT

- NASA Silver Group Achievement Award for JSOC flood recovery efforts 2025
- Stanford University Libraries Data Sharing Prize, Stanford Solar Observatories Group 2026
- Editor's Choice, *Solar Physics*, for *The Sun's Large-Scale Flows I* 2024
- Prepared and published the SWAN-SF benchmark dataset used in solar-flare AI work 2020

SELECTED PUBLICATIONS

Mahajan, S. S., Upton, L. A., Antia, H. M., Basu, S., DeRosa, M. L., Hess Webber, S. A., et al.
The Sun's Large-Scale Flows I: Measurements of Differential Rotation & Torsional Oscillation. *Solar Physics* 299, 38 2024

Mahajan, S. S., Sun, X., and Zhao, J.
Removal of Active Region Inflows Reveals a Weak Solar Cycle Scale Trend in the Near-Surface Meridional Flow. *Astrophysical Journal* 950, 63 2023

Dash, S., DeRosa, M. L., Dikpati, M., Sun, X., **Mahajan, S. S.**, Liu, Y., and Hoeksema, J. T.
Ensemble Kalman Filter Data Assimilation into the Surface Flux Transport Model to Infer Surface Flows. *Astrophysical Journal* 2024

SKILLS

Python, MATLAB, Fortran, bash, OpenMP, MPI, C++, CUDA, scientific computing, time-series analysis, machine learning workflows, data analysis, visualization, and technical writing.

EDUCATION

<i>Ph.D.</i> in Astronomy, Georgia State University	2019
<i>M.S.</i> in Physics, Georgia State University	2017
<i>M.Tech.</i> in Engineering Physics, IIT (BHU)	2014